

Western Mathematics

2023

YEAR 11
FINAL
EXAMINATION

Mathematics Standard

General Instructions

- Reading time – 10 minutes
- Working time – 2 hours
- Write using black pen
- Approved calculators may be used
- A reference sheet is provided at the back of this paper
- In Questions in Section II, show relevant mathematical reasoning and/or calculations

Total marks : **Section I – 10 marks** (pages 2 – 6)
80

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II – 70 marks (pages 7 – 29)

- Attempt Questions 11 – 35
- Allow about 1 hour and 45 minutes for this section

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1 – 10

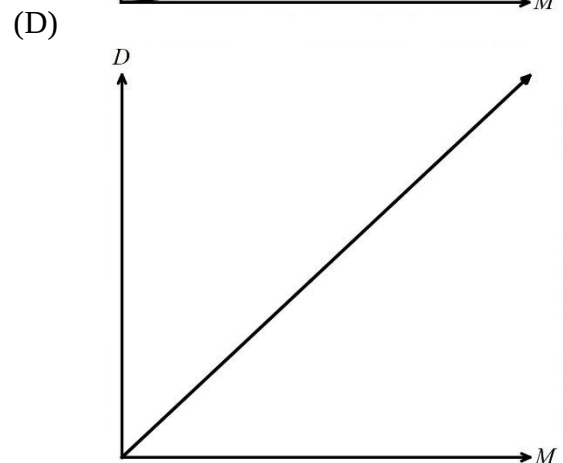
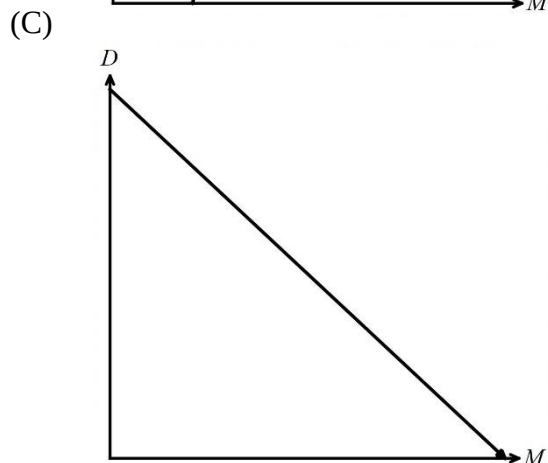
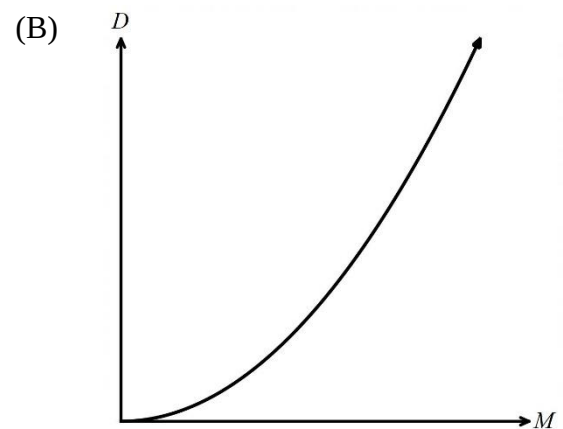
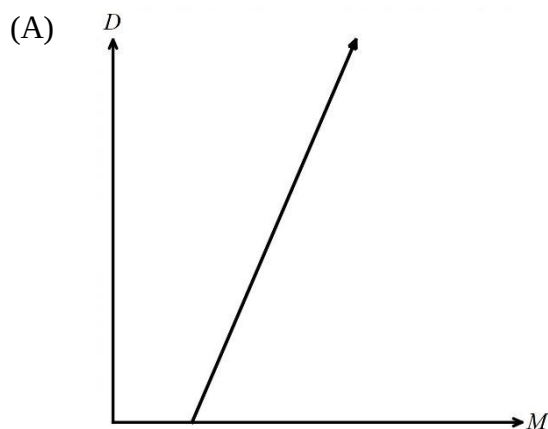
1. Liam times his sister Lilith completing one lap of an oval in one minute and twenty seconds, using an analogue watch which has markings every second.

What is the percentage error in his measurement?

- (A) 0.5%
- (B) 0.625%
- (C) 5.0%
- (D) 6.25%

2. For two quantities D and M , it is known that D is directly proportional to M .

Which graph could represent the relationship between D and M ?



3. All products that a company sells must have a GST added to the base price. Most products are offered at discounted prices. Two employees calculate sales. Aimee adds the GST to the base price first, then calculates the discounted price. Beatrix takes the discount off the base price first then adds the GST. Which statement is true?
- (A) Aimee will always get the same final price as Beatrix
- (B) Aimee will get the same final price as Beatrix on some products, but not all
- (C) Aimee will always get a higher final price than Beatrix
- (D) Beatrix will always get a higher final price than Aimee
4. Meghan's quarterly Water Account is shown below.

Your Water Account Itemised			
Service Charges for period 1 st July 2023 to 30 th Sept 2023			
	Number of Services	Charge	Amount
Sewer Service - Residential	1	\$120.40	\$120.40
Stormwater Drainage - Residential	1	\$35.60	\$35.60
Water Service - Residential	1	\$40.20	\$40.20
Total Service Charges			\$196.20
Usage Charges – Water Meter reading details over page			
	Usage (kL) or No. of charges	Tariff or Charge	Amount
Sewer (Wastewater) Usage - Residential	1	\$30.00	\$30.00
Water Usage (kL)	36	\$2.60	\$93.60
Total Usage Charges			\$123.60
Total amount due			\$319.80

Meghan wants to budget an amount from each weekly pay to cover her future water accounts. She allows for 45 kL of water usage every quarter in case there is extra usage or price rises. How much should she put aside each week?

- (A) \$15.99
- (B) \$24.60
- (C) \$25.40
- (D) \$26.40

Questions 5 and 6 refer to the following:

The back-to-back stem and leaf plot below compares the scores on a survey for members of two groups.

Group 1							Group 2					
						0	0	2	4	6		
						1	1	2	2	2	6	
9	6	5	3	3	0	2	2	3	6	6	9	
						3	1	3	3	5	7	9
						4	0	4	8			
						5	5	9				

5. Which of these statements is true in comparing the two groups?

- (A) Group 1 has a greater number of members
- (B) Group 1 has the greater median
- (C) Group 1 has the greater mode
- (D) Group 1 has the greater range

6. What is the interquartile range for Group 2?

- (A) 12
- (B) 26
- (C) 38
- (D) 59

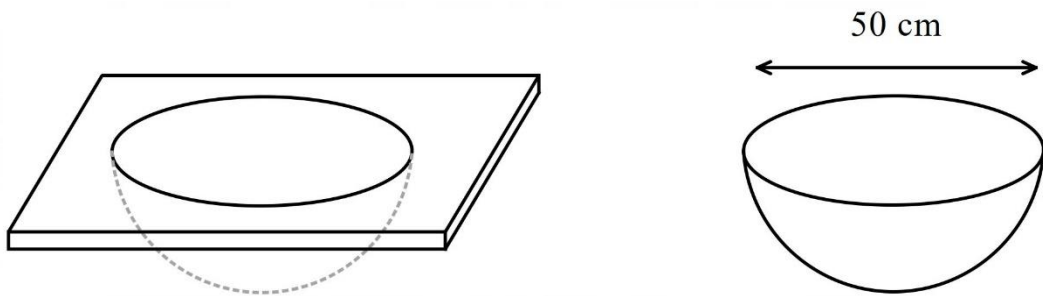
7. The four members of the K-Pop group BlackRose receive fan comments to their social media sites.

The number of fan comments received in a month is shown in the table below.

Group Member	Number of fan comments. (millions)
Seulgi	1.29
Candy	2.45
Giselle	1.46
Yuna	2.80

If a fan comment for the group is chosen at random, what is the probability that it isn't for Yuna?

- (A) 0.28
- (B) 0.35
- (C) 0.65
- (D) 5.20
8. The basin in a bathroom is hemispherical in shape and has a diameter of 50 cm.
- Given that $1\,000\text{ cm}^3$ holds 1 litre of water, what is the capacity of the basin, to the nearest 10^{th} of a litre?



- (A) 32.7 litres
- (B) 65.4 litres
- (C) 327.2 litres
- (D) 654.5 litres

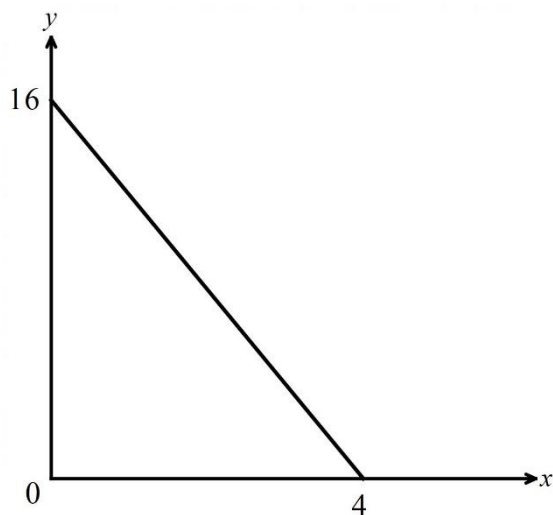
9. Fausia is a journalist and is paid a salary of \$81 900 per annum.

Over a four-week period she keeps track of her hours.

	Week 1	Week 2	Week 3	Week 4
Hours Worked	34	40	32	34

Based on the average hours that she worked over these four weeks, what is her hourly rate of pay?

- (A) \$39.38
- (B) \$45.00
- (C) \$46.32
- (D) \$49.22
10. What is the gradient of the line shown below?



- (A) -4
- (B) $-\frac{1}{4}$
- (C) $\frac{1}{4}$
- (D) 4

Western Mathematics

**2023 YEAR 11 FINAL
EXAMINATION**

Class and Teacher

Mathematics Standard

Student Number

Section II Answer Booklet

Student Name

70 marks

Attempt Questions 11 – 35

Allow about 1 hour and 45 minutes for this section

Instructions

- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.
- Your responses should include relevant mathematical reasoning and/or calculations.
- Extra writing space is provided at the back of the booklet.
If you use this space, clearly indicate which question you are answering.

Question 11 (3 marks)

The formula $s = ut + \frac{at^2}{2}$ is used in physics.

- (a) Use the formula to find the value of s when $a = 10$, $t = 4$ and $u = 12$.

1

.....

.....

.....

- (b) Use the formula to find the value of u when $a = 3$, $s = 84$ and $t = 6$.

2

.....

.....

.....

.....

.....

Question 12 (2 marks)

Light travels at an approximate speed of 300 000 km/second.

2

It takes 4.35 minutes for light to travel from the Sun to Mars.

Calculate the distance from the Sun to Mars. Give your answer in kilometres, written in scientific notation.

.....

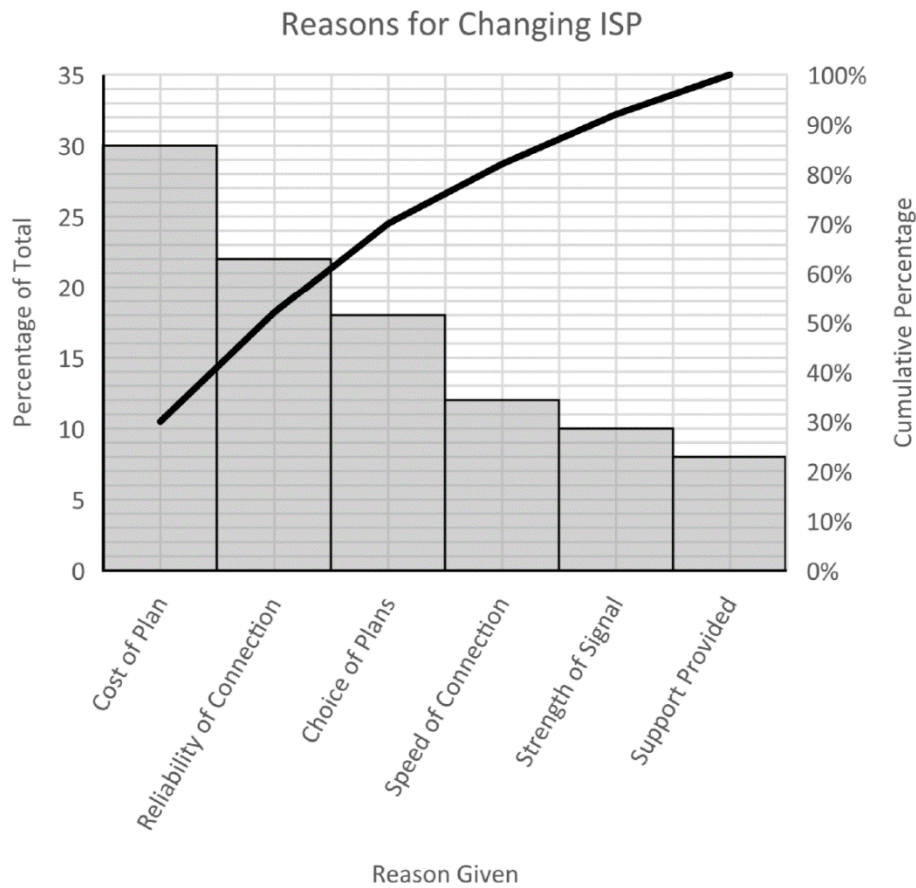
.....

.....

.....

Question 13 (2 marks)

Optel is an Internet Service Provider. When customers change to another provider, they are asked the main reason for their change. The Pareto Chart below summarises the reasons that people gave for changing to another provider.



- (a) If Optel initially wanted to make changes that would address 70% of the respondents, what is the least number of issues that they could address?

1

.....

.....

- (b) If Optel only addressed the technical issues (Reliability, Speed and Strength), what percentage of issues would it address?

1

.....

.....

.....

.....

Question 14 (2 marks)

Rearrange the formula $m = \frac{k^2 + wp}{d}$ to make p the subject.

2

.....

.....

.....

.....

.....

Question 15 (2 marks)

Marielle bought a used car in January 2022 for \$12 600.

The car is depreciating using the straight-line method, and in January 2023 it was valued at \$11 592.

She plans to sell it in January 2028 after she has completed her university course.

What will its value be when she sells the car?

2

.....

.....

.....

.....

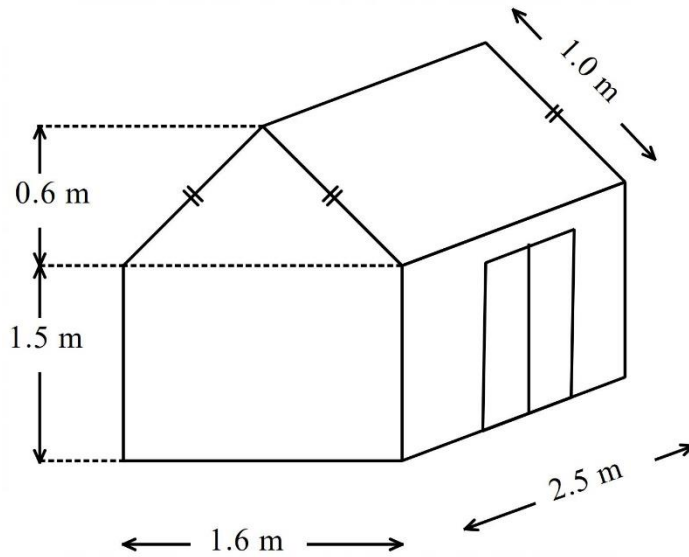
.....

.....

Question 16 (2 marks)

The small shed, shown below, has its walls, doors and roof made of sheet-metal and it is built on a concrete base.

2



Find the area of sheet-metal needed for the shed.

.....

.....

.....

.....

.....

.....

Question 17 (2 marks)

Greg and John both conduct a trial of tossing a biased four-sided die. Their results are shown in the table below.

2

Score	John's result	Greg's result
1	24	6
2	36	10
3	68	12
4	32	4
Total	160	32

Explain why one set of results would be better than the other in determining the probability that a “3” is rolled on the die and give that probability.

.....

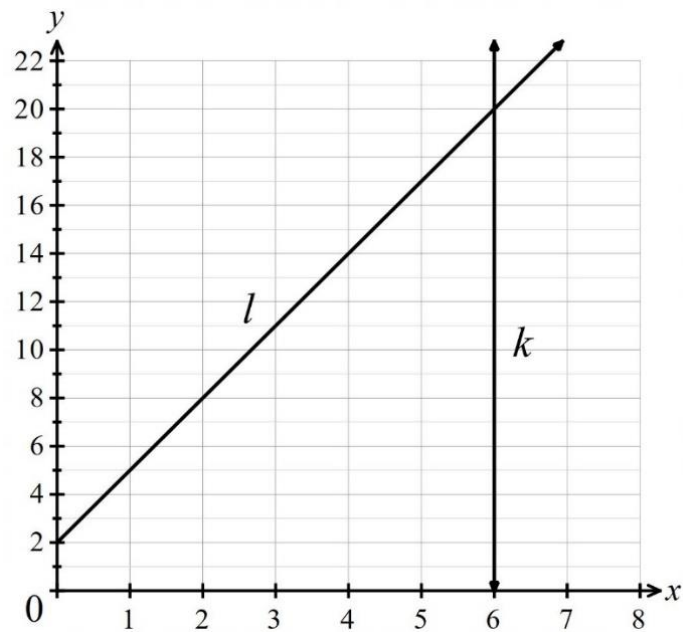
.....

.....

.....

Question 18 (3 marks)

Two straight line graphs are shown on the set of axes below, labelled k and l .



(a) Write the equation that describes line k . 1

.....

.....

(b) Write the equation that describes line l . 2

.....

.....

.....

.....

Question 19 (4 marks)

The frequency distribution table below summarises the test results of 80 students.

Marks	Class Centre x	Frequency f	fx	Cumulative Frequency
1 - 5	3	1	3	1
6 - 10	8	4	32	5
11 - 15	13	10	130	15
16 - 20	18	13	234	28
21 - 25	23	18	414	46
26 - 30	28	16	448	62
31 - 35	33	10	330	72
36 - 40	38	8	304	80

80

(a) What percentage of marks were greater than 20? 1

.....

.....

(b) Calculate the mean mark for the data. 1

.....

.....

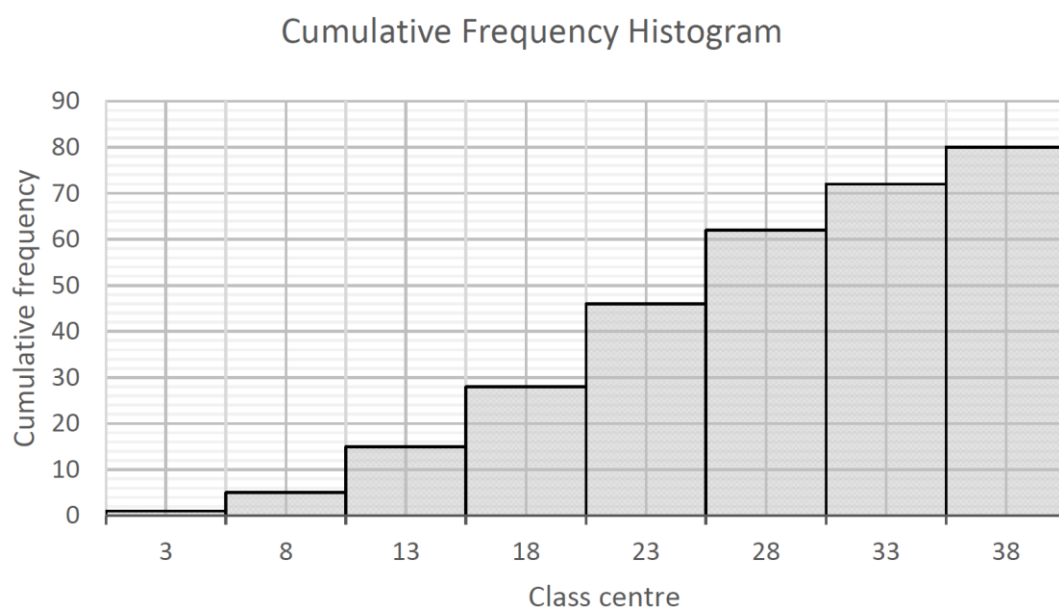
.....

Question 19 continues on page 15

Question 19 (continued)

- (c) A cumulative frequency histogram has been drawn for the data below. On the same axes, draw a cumulative frequency polygon, and use it to estimate the median of the data.

2



Question 20 (2 marks)

A wheelbarrow which weighs 4.5 kg will hold 35 bricks, each of which weighs 3.8 kg.

2

A platform on a crane has enough space to fit 15 loaded wheelbarrows, but it has a load limit of 1.5 tonnes.

How many full wheelbarrows of bricks could be loaded onto the platform on the crane, so that they could be safely lifted?

.....

.....

.....

.....

.....

Question 21 (2 marks)

A 50-gram egg is rated as having an energy rating of 78 Calories.

2

Sarah is advised to take no more than 2000 kilojoules of energy from eggs in a week, due to cholesterol issues.

Given that 1 kilojoule is equal to 0.24 Calories, how many eggs should Sarah limit herself to, in a week?

.....

.....

.....

.....

.....

.....

Question 22 (2 marks)

2

Michelle wants to buy a 2-year-old all-wheel drive sedan priced at \$46 400 and Simon wants a 5-year-old 4WD Ute priced at \$44 000.

Michelle has an offer of car insurance based on 5% of the value of the car while Simon has a quote which is 8% of the vehicle’s value.

Both will need to pay stamp duty on the sale of the vehicle, with the rates being shown below.

Motor vehicle value	Duty payable
Up to \$44,999	\$3 for every \$100
\$45,000 or more*	\$1,350 plus \$5 for every \$100 over \$45 000

What is the amount that each will pay to own and insure their vehicles?

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

Question 23 (3 marks)

Solve the equation $5g - 12 = 3(g + 6)$.

3

.....

.....

.....

.....

.....

.....

Question 24 (2 marks)

Yumi works a normal 35-hour week and is paid time-and-a-half for the first five hours of overtime and double time after that.

2

Yumi’s normal hourly rate is \$44.60/hour and her hours for a week are shown below.

Day	Mon	Tue	Wed	Thu	Fri	Sat
Hours	7	8	7	9	7	4

Calculate her pay for the week shown.

.....

.....

.....

.....

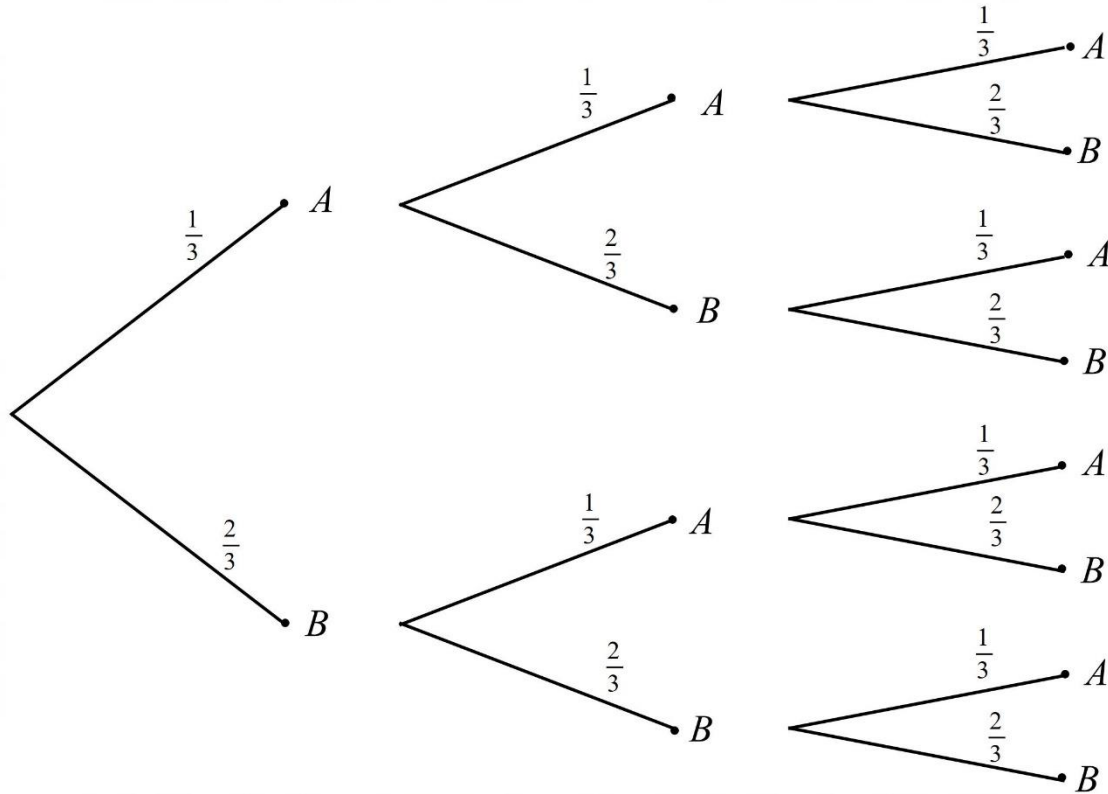
.....

Question 25 (3 marks)

A spinner has two options, A and B with the probability of B being $\frac{2}{3}$.

The spinner is spun three times and the result recorded.

The tree diagram below shows the outcomes.



(a) What is the probability that it lands on B three times?

1

.....

.....

(b) What is the probability that it lands on A twice and B once, in any order?

2

.....

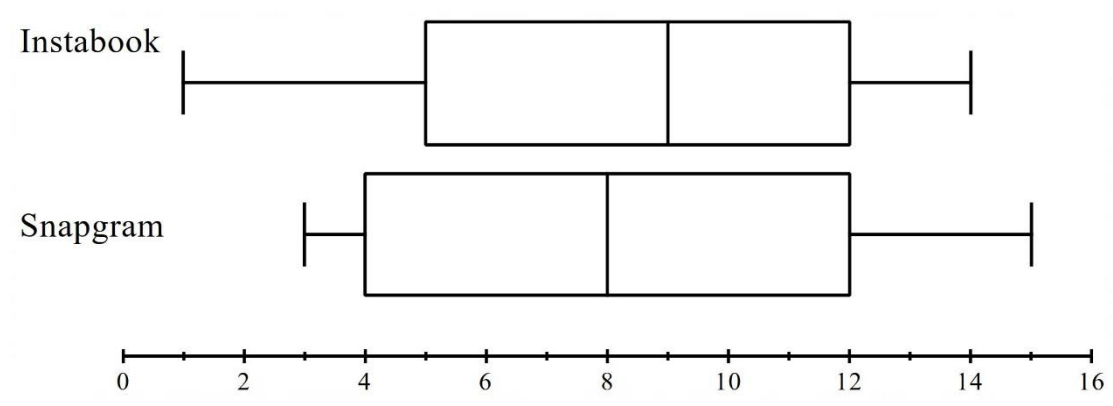
.....

.....

.....

Question 26 (3 marks)

Users of two social media sites are surveyed on the number of hours they are active on the sites per day.
The results are summarised in the parallel boxplots below.



- (a) Compare the medians of the two sets of data and determine which is greater. 1

.....
.....

- (b) Compare the range for the two sets of data and determine which is greater. 1

.....
.....

- (c) Compare the interquartile range for the two sets of data and determine which is greater. 1

.....
.....
.....

Question 27 (4 marks)

A car is travelling along a straight road at a constant speed of 90 km/h.

- (a) How many metres will the car travel in 2 minutes?

1

.....

.....

.....

.....

- (b) How many seconds would it take for the car to travel 100 metres.

1

.....

.....

.....

- (c) For this car, in the road conditions present, the stopping distance (D km) from when the brakes are applied is given by $D = 0.00001 s^2$, where s is the speed in km/h.

2

Find the stopping distance in metres.

.....

.....

.....

.....

.....

Question 28 (3 marks)

Josh has invested \$12 000 in an account that pays 4.8% per annum interest, compounded monthly.

- (a) Calculate the monthly interest rate.

1

.....

- (b) A table has been started to calculate the compound interest as it accumulates.
Complete the table for May and June.

1

Month	Principal at Start of Month	Interest earned	Principal at End of Month
January	\$12 000.00	\$48.00	\$12 048.00
February	\$12 048.00	\$48.19	\$12 096.19
March	\$12 096.19	\$48.38	\$12 144.57
April	\$12 144.57	\$48.58	\$12 193.15
May			
June			

.....

.....

.....

.....

- (c) Compare the amount of interest earned over the first six months using compound interest with the amount if simple interest was applied.

1

.....

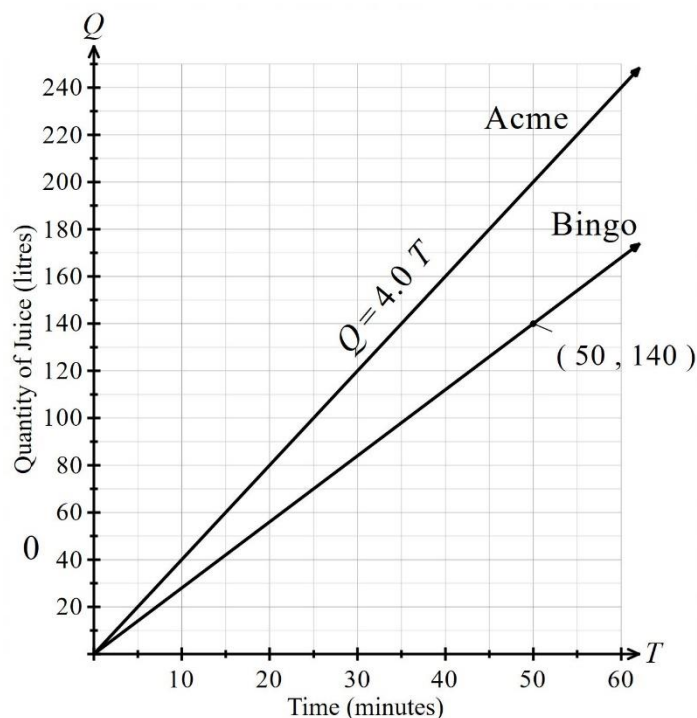
.....

.....

.....

Question 29 (3 marks)

Two lines are shown on the set of axes below, which represent the quantity of juice produced by two companies (Acme and Bingo) over an hour.



Both lines represent a direct variation between time T and quantity Q .

- (a) Write the equation which describes the variation for the Bingo company.

2

.....

.....

.....

- (b) How many more litres will Acme produce, compared to Bingo in an hour?

1

.....

.....

.....

Question 30 (3 marks)

The table below shows the income tax rates for Australian residents in the 2022–23 financial year.

Taxable income	Tax on this income
0 – \$18,200	Nil
\$18,201 – \$45,000	19c for each \$1 over \$18,200
\$45,001 – \$120,000	\$5,092 plus 32.5c for each \$1 over \$45,000
\$120,001 – \$180,000	\$29,467 plus 37c for each \$1 over \$120,000
\$180,001 and over	\$51,667 plus 45c for each \$1 over \$180,000

The above rates **do not** include the Medicare Levy of 2%.

Jared had an annual taxable income of \$50 240 for the 22-23 financial year and he has had \$310 deducted each fortnight and paid to the tax office.

- (a) Calculate the income tax that Jared is due to pay.

1

.....

.....

.....

.....

- (b) Calculate his Medicare levy and determine the amount of tax refund or bill that Jared will have after lodging his tax return.

2

.....

.....

.....

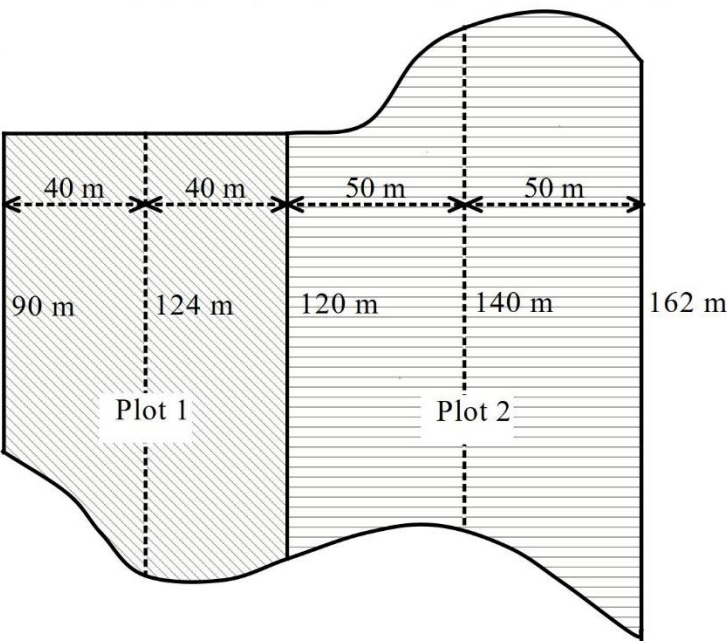
.....

.....

.....

Question 31 (3 marks)

Two plots of land, labelled Plot1 and Plot 2 below, are side by side on a road.



A real-estate agent says, “Plot 2 is about twice the size of Plot 1.”

A customer replies “No, I think it’s only about one and a half times as big.”

- (a) Use two applications of the trapezoidal rule to find the area of Plot 1. 1

.....

.....

.....

- (b) Use two applications of the trapezoidal rule to find the area of Plot 2 and determine which statement is more correct. 2

.....

.....

.....

.....

.....

Question 32 (3 marks)

The table below compares the percentage of first-time blood donors in Australia in each of the eight blood groups from 1994 and 2019.

Blood Type	O+	O-	A+	A-	B+	B-	AB+	AB-
Percentage from 1994	40.0	9.0	31.0	7.0	8.0	2.0	2.0	1.0
Percentage from 2019	37.8	8.7	29.6	5.4	12.4	1.6	3.9	0.6

- (a) One person's result is chosen at random from the 2109 data set. 1

What is the probability that the blood type will be AB (Either AB+ or AB-)?

.....

.....

.....

- (b) A person's results are chosen at random from each data set. 2

Compare the probabilities that the blood type will be positive (O+, A+, B+ or AB+) in each year?

.....

.....

.....

.....

Question 33 (3 marks)

A part of the train timetable on the line from Charleston to St George, is shown below.

Station	Arrival Times (AM)								
Charleston (Departure time)	3:00	3:30	4:00	4:30	5:00	5:30	6:00	6:30	7:00
Crystal Springs	3:25		4:25		5:25		6:25		7:25
Dingo Flat	3:39		4:39		5:39		6:39		7:39
Dry Lake	3:57		4:57		5:57		6:57		7:57
Arrowline	4:12	4:30	5:12	5:30	6:12	6:30	7:12	7:30	8:12
Boulder	4:42		5:42		6:42		7:42		8:42
Riverside	5:18		6:18		7:18		8:18		9:18
Bunkerville	5:30	5:42	6:30	6:42	7:30	7:42	8:30	8:42	9:30
Littlefield	5:54	6:04	6:54	7:04	7:54	8:04	8:54	9:04	9:54
Echo Ridge	6:21		7:21		8:21		9:21		10:21
St George	6:33	6:40	7:33	7:40	8:33	8:40	9:33	9:40	10:33

- (a) Edwina needs to travel from Dry Lake and reach St George station by 8:10 to catch a connecting bus. At what time should she catch the train from Dry Lake station?

1

.....

- (b) Will needs to travel from Charleston and reach Arrowline station in time to attend a 30-minute meeting which occurs 10 minutes from the station at 5:45. He then wants to catch a train to Echo Ridge as soon as possible. At what time should he catch the trains from the Charleston and Arrowline Stations?

2

.....
.....
.....

Question 34 (4 marks)

- (a) Cameron has a ticket to travel from Sydney, NSW (UTC+10) to Brasilia, Brazil (UTC−3). 1

He texts his friend Gabriel, who lives in Brasilia, at 5:00 pm, Sunday 3rd September, Sydney time.

At what local time does Gabriel receive the text?

.....

.....

.....

- (b) The text says that he will leave Sydney at 11:00 am on Tuesday 5th September and that the flight will take 18 and a half hours. 2

What local time in Brasilia will he arrive?

.....

.....

.....

.....

- (c) The distance from Sydney to Brasilia is 14 220 km. 1
- What is the average speed of the flight?

.....

.....

.....

Question 35 (5 marks)

Sienna has moved into a four-bedroom unit which she will share with three friends. She will pay a quarter of the rent and of the utilities (electricity and water) for the unit. She has just got her first full time job, which pays \$1 964.00 per fortnight after tax.

- (a) The rent on the unit is \$492.00 per week and the most recent quarterly bill for electricity was \$572.00 and for water was \$260.00. 2

She budgets for all these by increasing the total for rent and utilities by 20% to safeguard against possible increases.

How much should she budget from her **fortnightly** pay to cover her share of these expenses?

.....

.....

.....

.....

- (b) Sienna is also purchasing a car using a loan which has monthly payments of \$572.00. The car has weekly running expenses of \$48.00 and annual service, insurance, and registration expenses of \$ 1831.70. How much should she budget from her fortnightly pay to cover her car related expenses? 2

.....

.....

.....

.....

- (c) She pays \$145.00 per week toward household groceries and food and allows \$40 per day to buy lunch and other incidental daily expenses. 1

How much does she have left from her fortnightly pay to save for other major purchases and entertainment?

.....

.....

.....

.....

End of Examination

Section II Extra writing space

If you use this space, clearly indicate which question you are answering.

[illegible]

REFERENCE SHEET

Measurement**Limits of accuracy**

Absolute error = $\frac{1}{2} \times \text{precision}$

Upper bound = measurement + absolute error

Lower bound = measurement – absolute error

Length

$$l = \frac{\theta}{360} \times 2\pi r$$

Area

$$A = \frac{\theta}{360} \times \pi r^2$$

$$A = \frac{h}{2}(a + b)$$

$$A \approx \frac{h}{2}(d_f + d_l)$$

Surface area

$$A = 2\pi r^2 + 2\pi rh$$

$$A = 4\pi r^2$$

Volume

$$V = \frac{1}{3}Ah$$

$$V = \frac{4}{3}\pi r^3$$

Trigonometry

$$\sin A = \frac{\text{opp}}{\text{hyp}}, \quad \cos A = \frac{\text{adj}}{\text{hyp}}, \quad \tan A = \frac{\text{opp}}{\text{adj}}$$

$$A = \frac{1}{2}ab \sin C$$

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

Financial Mathematics

$$FV = PV(1 + r)^n$$

Straight-line method of depreciation

$$S = V_0 - Dn$$

Declining-balance method of depreciation

$$S = V_0(1 - r)^n$$

Statistical Analysis

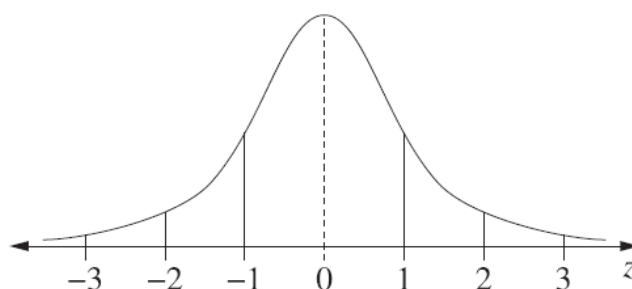
An outlier is a score

less than $Q_1 - 1.5 \times IQR$

or

more than $Q_3 + 1.5 \times IQR$

$$z = \frac{x - \mu}{\sigma}$$

Normal distribution

- approximately 68% of scores have z -scores between -1 and 1
- approximately 95% of scores have z -scores between -2 and 2
- approximately 99.7% of scores have z -scores between -3 and 3

Western Mathematics
2023 Year 11 Final Examination
Mathematics Standard

Name _____ Teacher _____

Section I – Multiple Choice Answer Sheet

Allow about 25 minutes for this section

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
 A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word **correct** and drawing an arrow as follows.

A ☒ B ☒ ^{correct} C ☐ D ☐

- | | | | | | | | | |
|-----|---|-----------------------|---|-----------------------|---|-----------------------|---|-----------------------|
| 1. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 2. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 3. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 4. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 5. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 6. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 7. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 8. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 9. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 10. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |